



Essential Grep

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A gentle introduction to searching and filtering files and logs. These questions are all done while logged in as "kali" and while in /home/kali.

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Question 1 using grep

Grep takes a minimum of 2 parameters. The first parameter is what you are looking for, and the second parameter is the file you are looking through. So to find all the lines containing "pendance" in /usr/share/dict/words you could say:

```
grep "pendanced" /usr/share/dict/words
```

Adapt this example to find all the words in /usr/share/dict/words and find the first word that contains the three letter sequence wpa

wpa word:

Tests: Ready

Dictionary search PASS

Question 2 count with grep

You can pass the output of grep directly into another command if you want to. This could allow you to nest searches (e.g. find all the words containing "wta" which end in "ing") or perform other sorts of processing. This is called piping, and the producer of the code goes on the left of the pipe character and the processing command goes to the right. The pipe character is often in the bottom right of the keyboard, and look like "|" (do not confuse it with the colon ":");

The wc command when given the option "-l" (minus and the letter L in lower case) counts how many lines it has been given. So

```
grep "danced" /usr/share/dict/words | wc -l
```

finds all lines containing the string "danced" and gives it to wc for counting.

Use grep piped through wc on file /usr/share/dict/words to find the number of words that contain the lower case letter x.

Number of words:

Tests: Ready

User: vinay mummadi (FULL)
Account: 40807459@live.napier.ac.uk
Registered Account

Account Links

Machine State: **RUN**

Refresh in: 0:18.

Boot progress: complete

Time left: Holding

control

Kali 2024-4

☐ fresh linux install image

☒ use previous image (if available)

Question 3 count with grep and save

You can save the output of a command to a file by ending the line with ">filename". So to save the number of lines to a file called "gordon" which contain "danced" in the dictionary you can do:

```
grep "danced" /usr/share/dict/words | wc -l > gordon
```

You can see file you have created using "ls" (a bit like "dir") and see the contents of a file using "cat" (e.g. cat gordon).

Use `grep` piped through `wc` on file `/usr/share/dict/words` to find the number of words that contain the letter `x` and save that answer to a file called `q1`.

```
(kali@host-3-89)-[~]
$ grep "wpa" /usr/share/dict/words
pawaw
pawaw's
pawaws
towpath
towpath's
towpaths
windowpane
windowpane's
windowpanes

(kali@host-3-89)-[~]
$ grep "danced" /usr/share/dict/words | wc -l
1

(kali@host-3-89)-[~]
$ grep "x" /usr/share/dict/words | wc -l
2209

(kali@host-3-89)-[~]
$ grep "danced" /usr/share/dict/words | wc -l > gordon

(kali@host-3-89)-[~]
$ cat gordon
1

(kali@host-3-89)-[~]
$ grep "x" /usr/share/dict/words | wc -l > gordon

(kali@host-3-89)-[~]
$ cat gordon
2209

(kali@host-3-89)-[~]
$ grep "x" /usr/share/dict/words | wc -l > q1

(kali@host-3-89)-[~]
$ cat q1
2209
```

Check Tests: Ready

x word count to file PASS

Question 4 negative grep

The `grep` command can take various options or flags. These are specified at the start of the command using a "-" sign. One useful flag is the "negation" search, which looks for lines which do not match. This flag is "-v". So to look for how many words DO NOT have "danced" in them, and save that to a file stuff, you could do

```
grep -v "danced" /usr/share/dict/words | wc -l > stuff
```

Use `grep` to find all lines in `/etc/passwd` that do **not** have `nologin` on the line. Send the output to file `nolog` (do this from the home directory of the kali user)

Check Tests: Ready

not nologin PASS

```
(kali@host-3-89)-[~]
$ grep -v "nologin" /etc/passwd > nolog

(kali@host-3-89)-[~]
$ cat nolog
root:x:0:0:root:/root:/usr/bin/zsh
sync:x:4:65534:sync:/bin:/bin/sync
dhcpd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpd:/bin/false
mysql:x:102:102:MariaDB Server,,,:/nonexistent:/bin/false
tss:x:103:103:TPM software stack,,,:/var/lib/tpm:/bin/false
Debian-snmpp:x:115:115::/var/lib/snmpp:/bin/false
postgres:x:117:118:PostgreSQL administrator,,,:/var/lib/postgresql:/bin/bash
speech-dispatcher:x:121:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
lightdm:x:127:128:Light Display Manager:/var/lib/lightdm:/bin/false
kali:x:1000:1000:Kali Linux,,,:/home/kali:/usr/bin/zsh

(kali@host-3-89)-[~]
$ wc -l nolog
10 nolog
```

Question 5 Download a log

Use the command `wget` to download one of our server's log files. You need to do:

```
wget http://linuxzoo.net/data/web.log -O log
```

This downloads one of our weblogs and saves it into a file called "log".

Check Tests: Ready

Downloaded ok PASS

Question 6 Any 404

Look for "file not found" errors in this weblog. This is error 404. Although not a perfect method, you can do this by searching for " 404 " in the log. The spaces are important, otherwise a search for

```
(kali@host-3-89)-[~]  
$ wget http://linuxzoo.net/data/web.log -O log  
--2026-02-27 20:07:02-- http://linuxzoo.net/data/web.log  
Resolving linuxzoo.net (linuxzoo.net) ... 146.176.166.1  
Connecting to linuxzoo.net (linuxzoo.net)|146.176.166.1|:80 ... connected.  
HTTP request sent, awaiting response ... 200 OK  
Length: 22619092 (22M) [text/plain]  
Saving to: 'log'
```

```
log 100%[=====]  
2026-02-27 20:07:02 (110 MB/s) - 'log' saved [22619092/22619092]
```

```
(kali@host-3-89)-[~]  
$ grep " 404 " log > notfound
```

```
(kali@host-3-89)-[~]  
$ cut -f1 -d" " notfound > ip
```

"404" would match "404hello" etc.

Once found save all of those to a file called "notfound".

Tests: Ready

Notfound lines PASS

Question 7 The IP numbers

Process the "notfound" file and save a list of only the IP numbers of each log entry. This can be done using

```
cut -f1 -d" " filename
```

This gives the first "field" of the file "filename", where fields are delimited using the space " " character. Save that info to a file called "ip".

Tests: Ready

Just IPs FAIL

Question 8 Duplicates

If you "cat ip" you will see that there are many duplicate IPs shown. The "sort -u filename" command will sort uniquely that file and remove duplicates. It also sorts the entry alpha numerically. What is the last IP printed if this uniqueness processing is applied to the ip file?

Last Unique IP:

Tests: Ready

Right IP PASS

```
(kali@host-3-89)-[~]
$ cat ip
10.1.12.169
10.1.12.169
10.1.12.169
10.1.12.169
10.1.12.169
10.1.12.169
10.1.12.169
10.1.12.169
```

Question 9 How many times

How many times does the above IP exist in the full log file "log"?

Count of Last Unique IP:

```
10.1.1.9.89
10.1.1.9.9
10.1.1.9.90
10.1.1.9.91
10.1.1.9.92
10.1.1.9.93
10.1.1.9.94
10.1.1.9.95
10.1.1.9.96
10.1.1.9.97
10.1.1.9.98
10.1.1.9.99

(kali@host-3-89)-[~]
$ grep "10.1.9.99" log | wc -l
3
```

Check

Tests: Ready

Right count PASS

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